

Geometric Measure Theory
A Beginner's Guide
Fourth Edition

Here as a child I watched my mom blow soap bubbles. My dad also encouraged all my interests. This book is dedicated to them with admiration.



Photograph courtesy of the Morgan family; taken by the author's grandfather, Dr. Charles W. Selemeyer

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A Beginner's Guide

Fourth Edition

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Front cover: John M. Sullivan's computer rendering of a soap bubble cluster obtained by area minimization on Ken Brakke's Evolver. See Preface, Chapter 13, and Section 16.8. © John M. Sullivan. All rights reserved.

Back cover: The author holds a student-commissioned stained glass window phase portrait of double bubbles in the 3D torus. See Section 19.7.

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Preface

Singular geometry governs the physical universe: soap bubble clusters meeting along singular curves, black holes, defects in materials, chaotic turbulence, crystal growth. The governing principle is often some kind of energy minimization. Geometric measure theory provides a general framework for understanding such minimal shapes, a priori allowing any imaginable singularity and then proving that only certain kinds of structures occur.

Jean Taylor used new tools of geometric measure theory to derive the singular structure of soap bubble clusters and sea creatures, recorded by J. Plateau over a century ago (see Section 13.9). R. Schoen and S.-T. Yau used minimal surfaces in their original proof of the positive mass conjecture in cosmology, recently extended to a proof of the Riemannian Penrose Conjecture by H. Bray. David Hoffman and his collaborators used modern computer technology to discover some of the first new complete embedded minimal surfaces in a hundred years (Figure 6.1.3), some of which look just like certain polymers. Other mathematicians are now investigating singular dynamics, such as crystal growth. New software computes crystals growing amidst swirling fluids and temperatures, as well as bubbles in equilibrium, as on the front cover of this book. (See Section 16.8.)

In 2000, Hutchings, Morgan, Ritoré, and Ros announced a proof of the Double Bubble Conjecture, which says that the familiar double soap bubble provides the least-area way to enclose and separate two given volumes of air. The planar case was proved by my 1990 Williams College NSF “SMALL” undergraduate research Geometry Group [Foisy *et al.*]. The case of equal volumes in $\mathbf{R}^3$ was proved by Hass, Hutchings, and Schlafly with the help of computers in 1995. The general $\mathbf{R}^3$ proof has now been generalized to $\mathbf{R}^n$ by Reichardt. There are partial results in spheres, tori, and Gauss space, an important example of a manifold with density (see Chapters 18 and 19).

This little book provides the newcomer or graduate student with an illustrated introduction to geometric measure theory: the basic ideas, terminology, and results. It developed from my one-semester course at MIT for graduate students with a semester of graduate real analysis behind them. I have included a few fundamental arguments and a superficial discussion of the regularity theory, but my goal is merely to introduce the subject and make the standard text, *Geometric Measure Theory* by H. Federer, more accessible.

Other good references include L. Simon’s *Lectures on Geometric Measure Theory*, E. Guisti’s *Minimal Surfaces and Functions of Bounded Variation*, R. Hardt and Simon’s *Seminar on Geometric Measure Theory*, Simon’s *Survey Lectures on Minimal Submanifolds*, J. C. C. Nitsche’s *Lectures on Minimal Surfaces* (now available in English), R. Osserman’s updated *Survey of Minimal*

Surfaces, H. B. Lawson's Lectures on Minimal Submanifolds, A. T. Fomenko's books on The Plateau Problem, and S. Krantz and H. Parks's Geometric Integration Theory. S. Hildebrandt and A. Tromba offer a beautiful popular gift book for your friends, reviewed by Morgan [14, 15]. J. Brothers and also Sullivan and Morgan assembled lists of open problems. There is an excellent Questions and Answers about Area Minimizing Surfaces and Geometric Measure Theory by F. Almgren [4], who also wrote a review [5] of the first edition of this book. The easiest starting place may be the Monthly article "What is a Surface?" [Morgan 24].

It was from Fred Almgren, whose geometric perspective this book attempts to capture and share, that I first learned geometric measure theory. I thank many graduate students for their interest and suggestions, especially Benny Cheng, Gary Lawlor, Robert McIntosh, Mohamed Messaoudene, and Marty Ross. I also thank typists Lisa Court, Louis Kevitt, and Marissa Barschdorf. Jim Bredt first illustrated an article of mine as a member of the staff of Link, a one-time MIT student newspaper. I feel very fortunate to have him with me again on this book. I am grateful for help from many friends, notably Tim Murdoch. Yoshi Giga and his students, who prepared the Japanese translation, and especially John M. Sullivan. I would like to thank my new editor, Lauren Schultz, and my original editor and friend Klaus Peters. A final thank you goes to all who contributed to this book at MIT, Rice, Stanford, and Williams. Some support was provided by National Science Foundation grants, by my Cecil and Ida Green Career Development Chair at MIT, and by my Dennis Meenan and Webster Atwell chairs at Williams.

This fourth edition includes updated material and references, recent results on planar soap films (Chapter 13), a new Chapter 18 on Manifolds with Density and Perelman's Proof of the Poincaré Conjecture and a new Chapter 19 on Double Bubbles in Spheres, Gauss Space, and Tori. Gauss space, defined as Euclidean space with Gaussian density, long studied by probabilists, appears along with general manifolds with density in Perelman's original 2003 paper on the Poincaré Conjecture. The proof of the Double Bubble Conjecture in spheres seems inextricably linked to its proof in Gauss space (see Chapter 19).

Bibliographic references are simply by author's name, sometimes with an identifying numeral or section reference in brackets. Following a useful practice of Nietzsche [2], the bibliography includes cross-references to each citation.

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